

Uncertainty Assessment of Electricity Price Forecasting Considering the Wind Onshore Generation

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Abstract—Following the deregulation of the electricity market, Electricity Price Forecasting (EPF) has become a critical task for stakeholders, including generation and retail companies. Traditional statistical models often fall short due to the complexity and non-linear relationships in electricity price data. To address this, we propose a hybrid CNN-BiGRU model that combines Convolutional Neural Networks (CNNs) and Bidirectional Gated Recurrent Units (BiGRUs) to capture spatial, temporal, and non-linear stochastic patterns in electricity prices, significantly improving forecasting accuracy. A robust data preprocessing pipeline is introduced, incorporating data cleaning, outlier detection using the Interquartile Range (IQR) method, and outlier treatment through Random Forest-based imputation. Feature selection is performed using the Least Absolute Shrinkage and Selection Operator (LASSO) to eliminate redundant features, reduce overfitting, and enhance the model's generalization ability. The proposed CNN-BiGRU model is rigorously evaluated against state-of-the-art benchmark models, demonstrating superior performance in terms of accuracy and robustness. Experimental results confirm its effectiveness as a reliable solution for short-term electricity price forecasting, particularly when considering Wind Onshore Generation, which further enhances forecasting accuracy. This highlights the importance of integrating renewable energy data into forecasting models, providing market participants with a valuable tool to navigate the challenges of price volatility and uncertainty in deregulated electricity markets.

Index Terms—Deep learning, Bidirectional long short-term memory, Sliding Window Approach, Recursive feature Elimination with Cross-Validation, Electricity price forecasting

I. INTRODUCTION

Electricity price forecasting, as a subset of energy forecasting combines knowledge from the electrical domain and econometrics to predict electricity prices (EP) [1]. This forecasting process depends on historical data on electricity prices and factors influencing those prices, such as demand, weather, generation, seasonality, etc. Earlier, the electricity market was typically governed by a single utility under government control. However, in the early 1990s, deregulation introduced a competitive market structure, starting with the UK and spreading globally [2]. As a result, electricity price forecasting became indispensable for generation companies, independent system operators (ISOs), and consumers [3]. Electricity mar-

kets present unique challenges for participants due to the inability to store electricity and the physical and reliability constraints on its transmission. These factors expose market participants to a variety of potential risks, that raise uncertainty in the system.

In recent decades, numerous models have been proposed for short-term (for several hours or one week ahead) electricity price forecasting. Artificial Intelligence has gained considerable attention compared to statistical models as it fails to capture the complex and non-linear pattern among the electricity price and its associated features, resulting in a less accurate model [4], [5]. That results in the shift of focus to Deep learning. Artificial Neural Networks (ANNs), inspired by the human brain's neural structure, have gained prominence in this domain. Among these, the Multilayer Perceptron (MLP) has emerged as a popular choice for Short-Term EPF (STEPF). In [6], the Recurrent neural network (RNN) model is used for forecasting electricity prices, but it struggles with the issue of exploding and vanishing gradients. LSTM utilizes sophisticated internal self-referencing cells to conserve long-term degradation trends and critical information simultaneously. Memory cells use an efficient gate mechanism to retain significant data and discard irrelevant details, effectively resolving problems like vanishing and exploding gradients [7]. In [8] [9], the LSTM model was used for price forecasting and gave better prediction than other deep-learning benchmark models as it excels in capturing temporal dependencies and nonlinear random behavior of electricity prices. For instance, [10] has utilized LSTM to integrate data from electricity markets, local weather, and calendar events and analyses long-term trends in hourly time series data to predict grid losses. Similarly, [11] proposed a Deep LSTM (DLSTM) model, which amalgamates multiple LSTM layers to address the limitations of shallow neural networks, particularly for long-interval time series forecasting. They further optimized the DLSTM architecture using Genetic Algorithms (GA). Another innovative approach was presented by [12], which has used a hybrid model combining Particle Swarm Optimization (PSO) with LSTM to enhance forecasting performance. Convolutional Neural Networks (CNNs) have also been explored

for STEPF. For example, [13] proposed a CNN model for price forecasting, employing Principal Component Analysis (PCA) for feature selection. Additionally, Gated Recurrent Units (GRUs) have demonstrated better performance in STEPF compared to other deep-learning models, as highlighted in [14].

The key contributions of this study are outlined as follows:

- 1) **Development of a CNN-BiGRU Model:** A novel hybrid model integrating Convolutional Neural Networks (CNNs) and Bidirectional Gated Recurrent Units (BiGRUs) is proposed for Short-Term Price Forecasting. This model leverages the spatial feature extraction capabilities of CNNs and the temporal dependency modeling of BiGRUs to effectively capture complex, nonlinear patterns and trends in electricity price data, significantly improving forecasting accuracy.
- 2) **Robust Data Preprocessing Framework:** An advanced data preprocessing pipeline is introduced, incorporating data cleaning, outlier detection using the Interquartile Range (IQR) method, and outlier treatment through Random Forest-based imputation. Additionally, feature selection is performed using the Least Absolute Shrinkage and Selection Operator (LASSO) technique to eliminate redundant features, mitigate overfitting, and enhance the model's generalization ability. This ensures the reliability and stability of the input data, further boosting forecasting performance.
- 3) **Comprehensive Model Validation:** The proposed CNN-BiGRU model is rigorously evaluated against state-of-the-art benchmark models under two scenarios: with and without considering Wind Onshore Generation. Experimental results demonstrate its superior performance in terms of accuracy and robustness, particularly when Wind Onshore Generation is incorporated, leading to significant improvements in forecasting precision. This establishes the CNN-BiGRU model as a highly effective and reliable solution for short-term electricity price forecasting, capable of addressing the challenges of price volatility and uncertainty in deregulated electricity markets.

The structure of this paper is as follows: In Section 2, the CNN-BiGRU architecture is discussed using the LASSO method as the approach. The case study that entails converting data into an appropriate format for model training is then presented in section 3. The discussed results are then displayed in section 4. The conclusion is covered in section 5.

II. METHODOLOGY USED

In this section, we have discussed the CNN-BiGRU architecture which is used for sequential time series forecasting of electricity prices, followed by the LASSO method for feature selection.

A. Convolution Neural Networks (CNN)

CNNs are widely applied in image processing due to their exceptional abilities in classification and regression tasks [15].

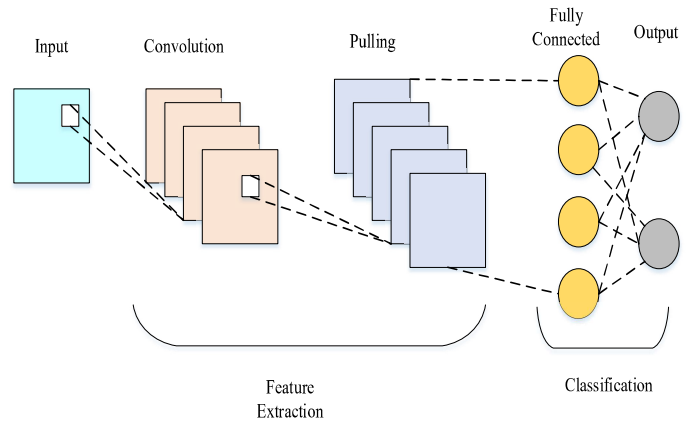


Fig. 1: CNN diagram

In essence, 1D-CNN utilize digital filters to extract temporal features from the time-domain representation of the electricity price series. This study focuses on analyzing and predicting one-dimensional time series data, specifically hourly electricity prices in Ontario. The convolution operations performed by 1D-CNN are visually represented in Fig. 1. As described mathematically in Eq. (1).

$$y_{t^1, c^1} = \sum_{n^1=1}^{c^1} \sum_{m^1=1}^{k^1} x_{t^1+m^1, n^1} * w_{m^1, n^1} + b_{c^1} \quad (1)$$

Let n^1 denote the channel index, m^1 represent the spatial position of the convolutional kernel, and k^1 denote the size of the convolutional kernel. The variable t corresponds to the input timestamp, while y_{t^1, c^1} represents the element of the output variables at timestamp t^1 and channel c^1 . The weight parameter and bias parameter of the convolution are denoted by w_{m^1, n^1} and b_{c^1} , respectively. When N^1 convolutional kernels are applied, the feature extraction process can be expressed in a simplified form as shown in Eq. (2):

$$Y_{B^1 \times T^1 \times N^1} = \text{stack}_1^{N^1} (W_{\text{Conv}} * X_{B^1 \times T^1 \times C^1} + b_{\text{Conv}}) \quad (2)$$

Here, $*$ represents the convolution operation, while $\text{stack}_1^{N^1}$ denotes the stacking of features extracted by N^1 convolutional kernels along the channel dimension. The parameters W^{Conv} and b^{Conv} correspond to the weight and bias matrices of the convolutional layer [16].

B. BiGRU

The GRU is an advanced variant of the traditional Recurrent Neural Network (RNN), designed to address the problem of vanishing gradients and improve long-term dependency modeling [17]. Unlike Long Short-Term Memory (LSTM) networks, which use three gates (input, forget, and output gates), GRU simplifies the architecture by employing only two gates: an update gate (Z_t) and a reset gate (r_t) as shown in the Fig. 2. The key steps in the GRU's forward propagation, including

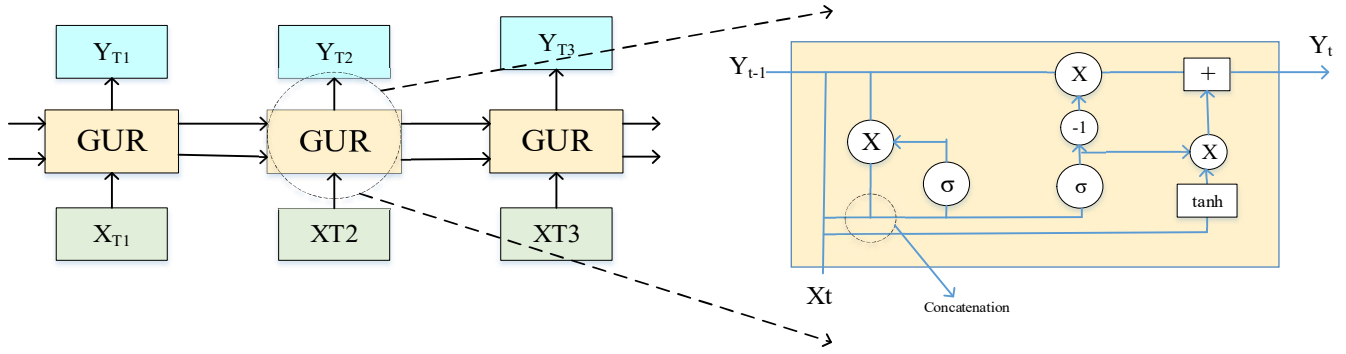


Fig. 2: GRU Architecture

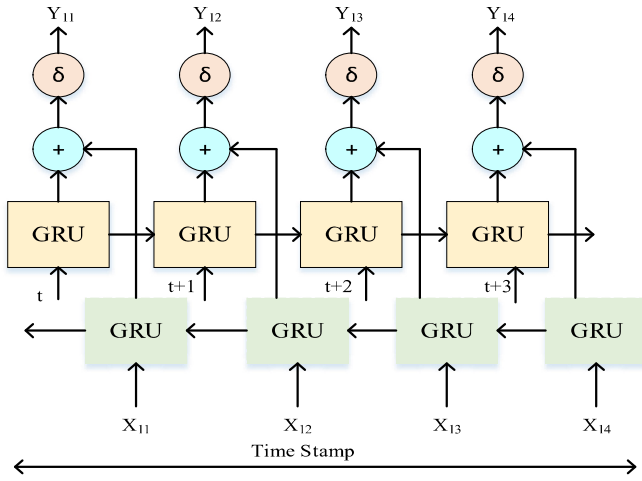


Fig. 3: Bidirectional GRU Architecture

the gating mechanisms and state updates, are represented in the mathematical Eq. (3) to Eq. (7) [18].

$$r_t = \sigma(W_r \cdot [h_{t-1}; x_t] + b_r) \quad (3)$$

$$z_t = \sigma(W_z \cdot [h_{t-1}; x_t] + b_z) \quad (4)$$

$$c_t = \tanh(W_c \cdot [r_t \odot h_{t-1}; x_t] + b_h) \quad (5)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot c_t \quad (6)$$

$$y_t = \sigma(W_o \cdot h_t) \quad (7)$$

The Update gate and reset gate equations are shown below in Eq. (5) and Eq. (6). The update gate determines how much historical information to retain, while the reset gate decides how much past information to discard. This streamlined structure makes GRU computationally faster and easier to train compared to LSTM, though it may sacrifice some flexibility and adaptability. GRU's ability to retain relevant information over time and discard irrelevant data makes it highly effective for sequential data tasks. Additionally, BiGRU models enhance performance by processing data in both forward and backward directions, allowing the model to leverage context from past and future states simultaneously as shown in the Fig. 3.

This bidirectional approach improves information acquisition, making the model more robust and reliable for tasks requiring comprehensive temporal understanding [18].

C. Least Absolute Shrinkage and Selection Operator (LASSO)

LASSO is a regression analysis technique that helps to identify the most significant features of a specific predictive task. It achieves this by penalizing the regression coefficients using the L1 norm, encouraging sparsity in the coefficient values. In the context of feature selection, LASSO assigns small or zero coefficients to irrelevant or less important features, effectively eliminating them from the model as shown in Eq. 8-9. By doing so, it helps to simplify the model and improve its interpretability while maintaining good predictive performance.

$$F_s = \text{Min}(\text{Arg RSS}) + \lambda \|F\| \quad (8)$$

$$RSS = \sum (P_{d,h} - \widehat{P}_{d,h})^2 \quad (9)$$

where RSS is the residual sum of squares, $P_{d,h}$ and $\widehat{P}_{d,h}$ is the actual and predicted values of EP. By selecting the relevant features, the LASSO method helps to enhance the accuracy and efficiency of the forecasting model by reducing noise and unnecessary complexity [19].

III. CASE STUDY

To validate the prediction accuracy of the proposed model, this study utilizes four years (2015–2018) of electricity price data from the Spanish Electricity Market, along with net load, and generation from conventional and non-conventional sources obtained from the Red Eléctrica de España portal and Spanish Transmission System Operator (TSO) [20]. Figure 4 illustrates the highly volatile nature of electricity prices over the four-year period (2015–2018).

A. Data Pre-processing

Data Pre-processing is crucial for enhancing the accuracy of the proposed model [21], in this section we have covered the data Pre-processing in three steps: 1) Data cleaning, where the treatment of the missing value is done, 2) feature selection, LASSO method of the feature section is adopted to eliminate the redundant features in the dataset as discussed in the II

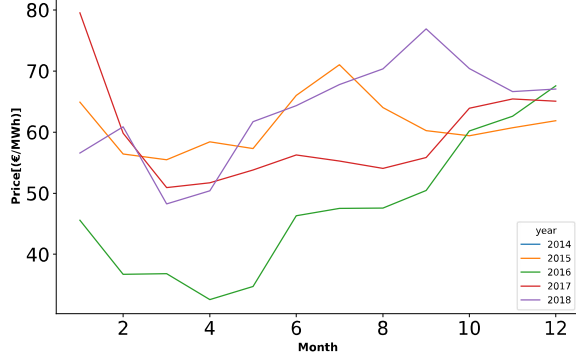


Fig. 4: Price distribution along the time period

section, which can be seen in the Fig. 6. 3) outlier treatment is done in this process, where the outlier are determined by interquartile range (IQR) method and replaced by its individual features medium [22]. Fig. 5, shows the (a) Boxplot for the generation from wind onshore before outlier detection; (b) after the outlier is replaced by forecasted values generated by a Random Forest model.

B. Performance Metrics

Mean absolute error (MAE), root mean square error (RMSE), symmetric mean absolute percentage error (sMAPE), and coefficient of determination (R2 Score) are the five performance metrics used in this study to validate the outcomes of the suggested model. These metrics are deduced from the mathematical Equation (10)–(13) [23], [24].

$$MAE = \frac{1}{24N_m} \sum_{d=1}^{N_m} \sum_{h=1}^{24} |P_{d,h} - F_{d,h}| \quad (10)$$

$$RMSE = \sqrt{\frac{1}{24N_m} \sum_{d=1}^{N_m} \sum_{h=1}^{24} |P_{d,h} - F_{d,h}|^2} \quad (11)$$

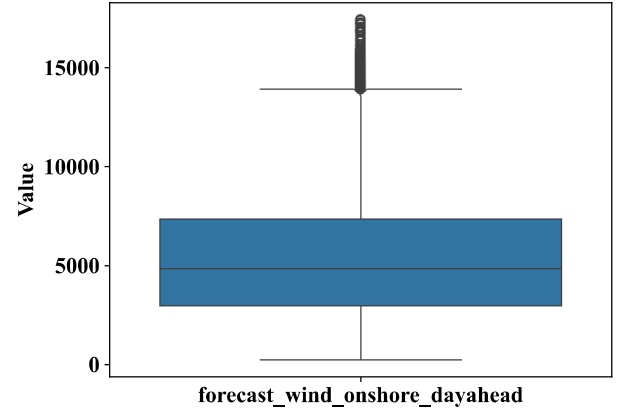
$$sMAPE = \frac{1}{24N_m} \sum_{d=1}^{N_m} \sum_{h=1}^{24} \frac{|P_{d,h} - F_{d,h}|}{|P_{d,h}| + |F_{d,h}|} \quad (12)$$

$$R^2 = 1 - \frac{\sum_{d=1}^{N_m} \sum_{h=1}^{24} |P_{d,h} - F_{d,h}|^2}{\sum_{d=1}^{N_m} \sum_{h=1}^{24} |P_{d,h} - \bar{P}_{d,h}|^2} \quad (13)$$

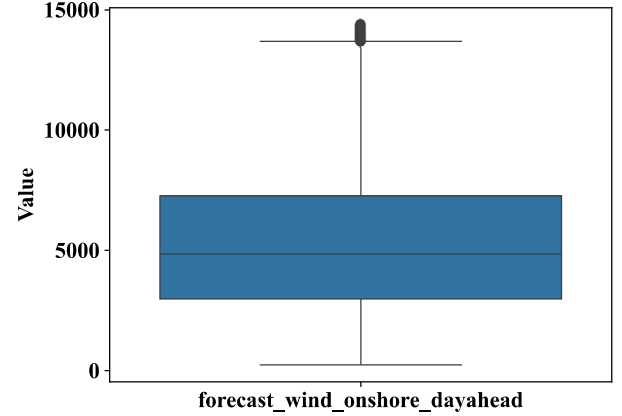
Where $P_{d,h}$, $F_{d,h}$ and $\bar{P}_{d,h}$ is real, forecast and mean price on day d and hour h ; N_m is the number of days.

C. Experimental Setting

In this study, the performance of the newly developed model is evaluated using a Spanish dataset. The dataset is partitioned into three subsets: training, validation, and testing, with a distribution of 60%, 20%, and 20%, respectively. The implementation of the model leverages several Python



(a)



(b)

Fig. 5: (a) Boxplot for the generation from wind onshore before outlier detection; (b) after the outlier is replaced by forecasted values generated by a Random Forest model.

libraries, including Numpy, Scikit-Learn, TensorFlow, Pandas, Matplotlib, and Seaborn, and is executed using Python 3.9.0. The computational setup consists of a 12th Generation Intel(R) Core(TM) i7-12700 processor running at 2.10 GHz, 16 GB of RAM, a 64-bit operating system, and an x64-based architecture.

IV. RESULTS AND DISCUSSION

In this study, an experimental comparison analysis is used to demonstrate the efficiency of the suggested paradigm. The analysis focuses on test data collected for one week from April 24 to April 30, 2018. The findings show that the suggested model performs better than a number of benchmark models. In addition, BiLSTM, LSTM, CNN, RF, GB, XGB, and MLP are the benchmark models used for comparison.

A. Electricity Price Forecasting Results for the Spanish Electricity Market

The Proposed Model demonstrates consistent improvements over the standard BiLSTM and other benchmark models in both scenarios—with and without considering Wind Onshore

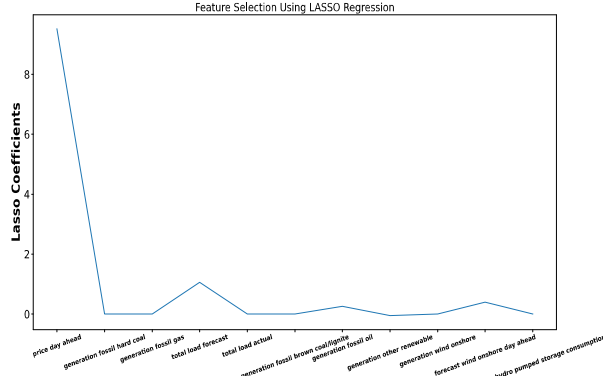


Fig. 6: Feature Selection Using LASSO Regression

TABLE I: Metrics for evaluating performance for the one-week test period (2018-04-24 to 2018-04-30) without taking wind onshore generation into account

Models	RMSE	MSE	MAE	MAPE	sMAPE	R2 Score
Proposed-Model	2.8801	8.2952	2.2419	0.0371	3.70	0.9382
BiLSTM	3.1145	9.7008	2.4651	0.0407	4.07	0.9280
LSTM	3.2496	10.5592	2.5246	0.0414	4.18	0.9165
CNN	4.9669	24.6698	3.6962	0.0617	6.56	0.8200
RF	18.4808	341.54	0.3174	0.2646	26.52	-0.8800
GB	18.5996	345.94	0.3188	0.2665	26.70	-0.9100
xgboost	19.2024	368.73	0.3255	0.2719	27.45	-1.0300
MLP	19.1592	367.08	0.3269	0.2739	27.57	-1.0260

Generation. Without Wind Onshore Generation, it achieves a 7.5% reduction in RMSE, 8.2% in MSE, and 9.0% in MAE, along with improved MAPE, sMAPE, and R2 scores as shown in Table I. When Wind Onshore Generation is included, the Proposed Model shows even better performance, with significant reductions in RMSE (11.6%), MSE (21.9%), and MAE (15.5%), scores as shown in Table II. These improvements highlight the model's superior accuracy and reliability in short-term electricity price forecasting. The inclusion of Wind Onshore Generation enhances all models' performance, particularly the Proposed Model, which closely follows actual price trends, as illustrated in Fig. 7.

B. Confidence Interval

For a 95% confidence interval (CI), it means that we are 95% confident that the value of the EP lies within the calculated interval. Similarly, for a 98% CI, we are 98% confident that the value of the EP is within the interval [25]. In Figure 8, the red line shows the predicted price and the purple area indicates 95% and 98% confidence that the predicted EP remains fixed in that area, respectively, where CI is calculated from the mathematical Eq. 14-15.

$$CI(95\%) = \mu \pm (1.96 * \sigma) \quad (14)$$

TABLE II: Metrics for evaluating performance for the one-week test period (2018-04-24 to 2018-04-30) with taking wind onshore generation into account

Models	RMSE	MSE	MAE	MAPE	sMAPE	R2 Score
Proposed-Model	2.5458	6.4812	1.8944	0.0318	3.18	0.9517
BiLSTM	2.7535	7.5822	2.0838	0.0349	3.49	0.9430
LSTM	2.8725	8.2503	2.1338	0.0354	3.59	0.9330
CNN	4.3906	19.2764	3.1235	0.0529	5.62	0.8440
RF	16.3347	266.82	0.2684	0.2236	22.52	-0.8800
GB	16.4385	270.28	0.2695	0.2252	22.68	-0.9100
xgboost	16.9738	288.12	0.2750	0.2298	23.35	-1.0300
MLP	16.9348	286.86	0.2762	0.2316	23.46	-1.0260

$$CI(99\%) = \mu \pm (2.33 * \sigma) \quad (15)$$

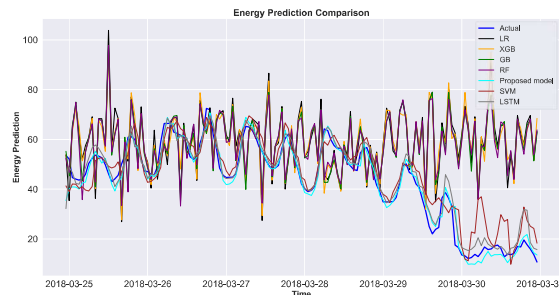
Where σ is the EP's standard deviation divided by the EP size's square root and μ is the mean of the EP.

CONCLUSION

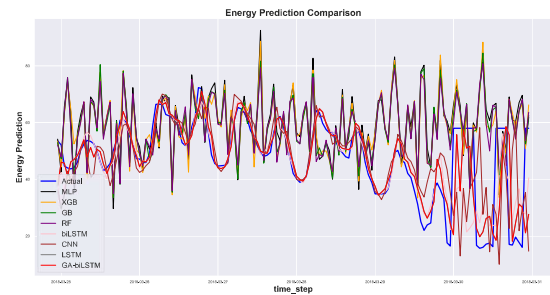
The proposed CNN-BiGRU model offers a robust solution for short-term electricity price forecasting by combining CNNs and BiGRUs to capture complex, non-linear patterns in price data. Enhanced by a rigorous data preprocessing pipeline and LASSO-based feature selection, the model demonstrates superior accuracy and reliability compared to benchmark models. Additionally, the study highlights the impact of uncertainty, showing that incorporating Wind Onshore Generation significantly improves forecasting performance. These findings underscore the importance of advanced preprocessing, hybrid deep learning architectures, and integrating renewable energy data, providing market participants with a reliable tool to address price volatility in deregulated electricity markets.

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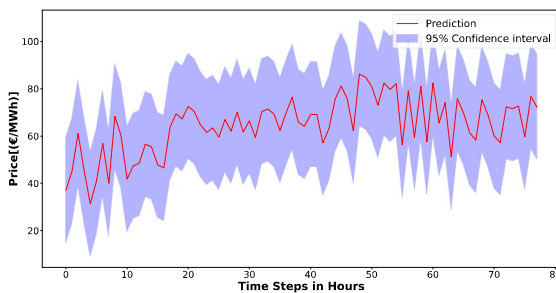


(a) Forecasting outcomes of the suggested model acquired for testing horizon without taking into account the Spanish market for wind onshore generation

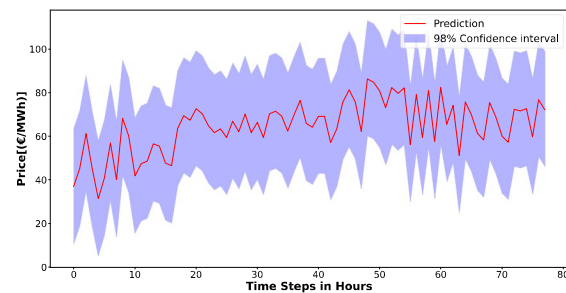


(b) Forecasting outcomes of the suggested model acquired for testing horizon with taking into account the Spanish market for wind onshore generation

Fig. 7: Forecasting results obtained for Spain Electricity Market



(a) Forecasting results obtained for 95 percent confidence interval in Spain Market



(b) Forecasting results obtained for 98 percent confidence interval in Spain Market

Fig. 8: Forecasting results obtained for different confidence interval in Spain Market

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